Optimal facility location

Choosing sensor stations to serve every region

Total cost = activation costs + service costs

Decision: can we serve every region within budget B?
Optimization: what is the cheapest complete solution?

Input: possible sites, regions, costs, and a budget.
Answer: open sites and one assigned site for each region.

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Example 1: one station wins

Required regions: North, Central, South Budget: 6

Site	Activation	North	Central	South
A	3	0	0	9
B	4	9	0	0
C	6	0	0	0

Region columns show extra service cost. All costs use fictional units.

Open C. Assign all three regions to C.

Total = 6 + 0 + 0 + 0 = 6 YES

Example 2: two stations win

C now costs 8 to activate. The budget is now 7.

Site	Activation	North	Central	South
A	3	0	0	9
B	4	9	0	0
C	8	0	0	0

The service costs and required regions stay the same.

Open A + B. A serves North/Central. B serves South.

Total = 3 + 4 + 0 + 0 + 0 = 7 YES

Checking a candidate answer

1. Check valid site names and one assignment per region.
2. Confirm every assigned station is open.
3. Add activation and service costs. Compare with budget.

Example 2: budget 7

A + B, assignments [A, A, B]: 7 + 0 = 7 ACCEPT

A alone, assignments [A, A, A]: 3 + 9 = 12 REJECT

Checking a supplied answer takes O(s + r) operations.

Acceptance establishes feasibility. Optimality needs more work.

Brute force tries every subset

Each station is open or closed: 2 choices per site.

2^s subsets

3 sites: 8 subsets
30 sites: 1,073,741,824 subsets

For A, B, C: none, A, B, C, AB, AC, BC, ABC

For each nonempty subset, assign every region to its cheapest open station. Add service and activation costs.

Work per subset: O(sr + s)
Worst-case total: O(2^s × (sr + s))